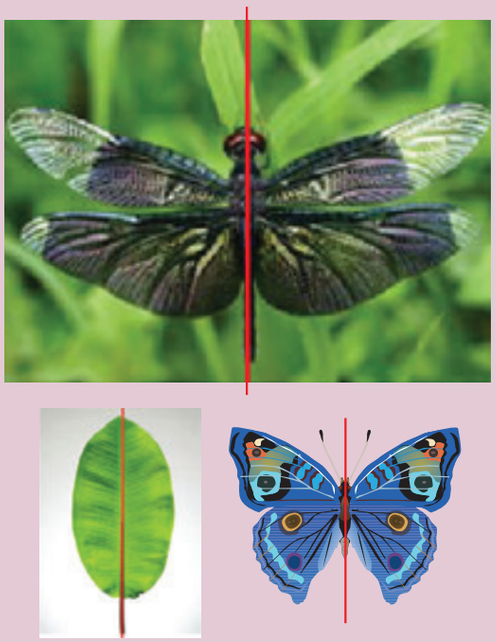
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the following question, "when the time varying magnetic field produces an electric field, why not the time varying electric field produce a magnetic field?"



Symmetry in nature

Later he proved that indeed it exists, which is often known as Maxwell's law of induction. In 1888, H. Hertz experimentally verified Maxwell's predication and hence, this understanding resulted in new technological invention, especially in wireless communication, LASER (Light Amplification by Stimulated Emission of Radiation) technology, RADAR (**Ra**dio **D**etection **A**nd **R**anging), etc.

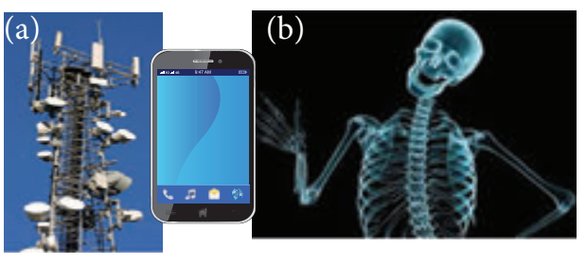


Figure 5.2 (a) cell phone and cell phone tower (b) X-ray radiograph of a human being

In today's digital world, cell phones (Figure 5.2 (a)) have greater influence in our day to day life. It is a faster and more effective mode of transferring information from one place to another. It works on the basis that light is an electromagnetic wave. In hospitals, the location of bone fracture can be detected using X-rays as shown in Figure 5.2 (b), which is also an electromagnetic wave. For cooking microwave oven is used. The microwave is also an electromagnetic wave. There are plenty of applications of electromagnetic waves in engineering, medical (example LASER surgery, etc), defence (example, RADAR signals) and also in fundamental scientific research. In this unit, basics of electromagnetic waves are covered.

5.1.1 Displacement current and Maxwell's correction to Ampere's circuital law

In unit 4, we studied Faraday's law of electromagnetic induction which states that the change in magnetic field produces an electric field. Mathematically

$$\oint_l \vec{E}\cdot\vec{dl} = -\frac{\partial}{\partial t}\Phi_B = -\frac{\partial}{\partial t}\oint_S \vec{B}\cdot\vec{dS} \qquad (5.1)$$

$$\underbrace{\oint_l \vec{E}\cdot\vec{dl}}_{\text{Electric field induced along a closed loop}} = \underbrace{-\frac{\partial}{\partial t}\Phi_B}_{\text{Variation of magnetic flux with time}} = \underbrace{-\frac{\partial}{\partial t}\oint_S \vec{B}\cdot\vec{dS}}_{\text{Changing magnetic flux } \Phi_B \text{ in the region enclosed by the loop}}$$

where Φ_B is the magnetic flux and $\frac{\partial}{\partial t}$ is the partial derivative with respect to time. Equation (5.1) means that the electric field $\vec{E}$ is induced along a closed loop by the changing magnetic flux Φ_B in the region encircled by the loop. Now the question asked by James Clerk Maxwell is 'Is converse of this statement true?' Answer is 'yes'. He

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